# Cauchy Integral Formula

# An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

(Dated: October 2026)

Abstract

A complete authored walkthrough of cauchy integral formula, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**CA-01 / Complex Analysis / Report v1**

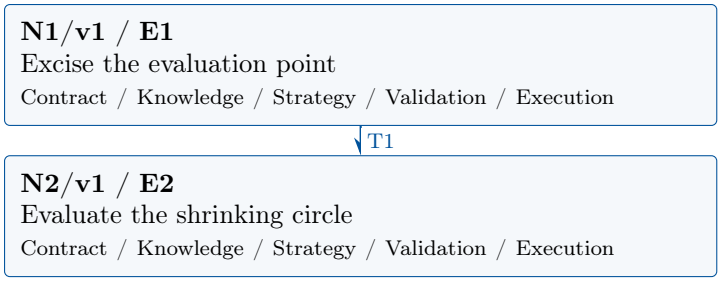
## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *Let $f$ be holomorphic on an open neighborhood of the closed disk $\overline{D(z_0, R)}$. For $w$ inside the disk, prove $f(w) = \frac{1}{2\pi i} \oint_{|z-z_0|=R} f(z)/(z-w)\, dz$, with positive orientation.*

### A. Foundations and permitted imports

Holomorphic means complex differentiable on an open neighborhood of the closed disk. Import Cauchy–Goursat in a finitely connected region with piecewise smooth boundary: the positively oriented boundary integral of a holomorphic function is zero. Also use continuity and the parametrization of a circle. Inner boundary components carry clockwise induced orientation.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1: N1/E1/v1 to N2.** The outer integral equals the counterclockwise inner integral.

**Receiving activity.** N2: Evaluate the shrinking circle

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: Equality of integrals; continuity at w; prior exports E1/v1

## III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

### A. N1/v1: Excise the evaluation point

**Object and role.** Move the integral to a small circle without crossing a singularity.

**Inputs.** w inside disk; f holomorphic near closure; foundational imports stated in §1

**C — Contract**

Own E1/v1: The outer integral equals the counterclockwise inner integral.

**K — Knowledge**

Retain w inside disk; f holomorphic near closure; foundational imports stated in §1; keep their assumptions and versions attached.

**S — Strategy**

Move the integral to a small circle without crossing a singularity.

**V — Validation**

The singular point must be removed; the sign of the inner boundary matters.

**E — Execution**

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N1.1 Choose $\rho > 0$ so the closed disk about w of radius $\rho$ lies inside $D(z_0, R)$.

N1.2 On the region between the outer circle and this small disk, $h(z) = f(z)/(z-w)$ is holomorphic.

N1.3 Cauchy–Goursat with its induced boundary orientations gives $\oint$ outer h $dz - \oint |z - w| = \rho$ h $dz = 0$, where both displayed circles are individually counterclockwise.

**Export E1/v1.** The outer integral equals the counterclockwise inner integral.

**Acceptance gate.** The singular point must be removed; the sign of the inner boundary matters. Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N1: The singular point must be removed; the sign of the inner boundary matters. Recheck dependent nodes N2.

### B. N2/v1: Evaluate the shrinking circle

**Object and role.** Subtract the constant value and estimate the remainder.

**Inputs.** Equality of integrals; continuity at w; prior exports E1/v1

**C — Contract**

Own E2/v1: $f(w) = (2\pi i)^{-1}$ times the given boundary integral.

**K — Knowledge**

Retain Equality of integrals; continuity at w; prior exports E1/v1; keep their assumptions and versions attached.

**S — Strategy**

Subtract the constant value and estimate the remainder.

**V — Validation**

An integral bound uses $circumference 2\pi\rho$ and denominator $\rho$; they cancel.

**E — Execution**

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N2.1 Split $f(z) = f(w) + (f(z) - f(w))$.

N2.2 The constant part integrates to $2\pi i f(w)$, since $z = w + \rho e^{it}$ gives $dz/(z-w) = i$ dt.

N2.3 The remainder satisfies $|\oint (f(z) - f(w))/(z-w)\, dz| \le 2\pi \sup_{|} z - w| = \rho |f(z) - f(w)| \to 0$ by continuity.

N2.4 The outer integral is independent of $\rho$, so taking $\rho \to 0$ gives $\oint$ outer $f(z)/(z-w)dz = 2\pi i f(w)$, the desired formula.

**Export E2/v1.** $f(w) = (2\pi i)^{-1}$ times the given boundary integral.

**Acceptance gate.** An integral bound uses $circumference 2\pi\rho$ and denominator $\rho$; they cancel. Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N2: An integral bound uses $circumference 2\pi\rho$ and denominator $\rho$; they cancel. Withhold this terminal result until its gate passes.

## IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

### A. §4.N1 — Excise the evaluation point

1. Choose $\rho > 0$ so the closed disk about w of radius $\rho$ lies inside $D(z_0, R)$.
2. On the region between the outer circle and this small disk, $h(z) = f(z)/(z - w)$ is holomorphic.
3. Cauchy–Goursat with its induced boundary orientations gives $\oint$ outer h $dz - \oint |z-w| = \rho$ h $dz = 0$, where both displayed circles are individually counterclockwise.

### B. §4.N2 — Evaluate the shrinking circle

1. Split $f(z) = f(w) + (f(z) - f(w))$.
2. The constant part integrates to $2\pi i f(w)$, since $z = w + \rho e^{it}$ gives $dz/(z - w) = i$ dt.
3. The remainder satisfies $|\oint (f(z)-f(w))/(z-w)\, dz| \leq 2\pi \sup_{|} z-w| = \rho|f(z)-f(w)| \to 0$ by continuity.
4. The outer integral is independent of $\rho$, so taking $\rho \to 0$ gives $\oint$ outer $f(z)/(z-w)dz = 2\pi i f(w)$, the desired formula.

□

## V. WORKED ORCHESTRATION

For $f(z) = z^2 + 1$, $z_0 = 0, R = 2, w = 1$, the formula predicts $\oint (z^2+1)/(z-1)dz = 4\pi i$. Algebraically $z^2 + 1 = (z - 1)(z + 1) + 2$; the polynomial term integrates to 0 and $2/(z - 1)$ integrates to $4\pi i$.

### A. Boundary and hypothesis audit

For w on the boundary the integrand is singular on the contour, and the excision argument is unavailable. Validation rejects the interior formula there; boundary limiting theorems require a different contract. Reversing the contour gives the negative of the stated integral.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

### B. Hypothesis boundary

For w on the boundary the integrand is singular on the contour, and the excision argument is unavailable. Validation rejects the interior formula there; boundary limiting theorems require a different contract. Reversing the contour gives the negative of the stated integral.

### C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.**

- Harvard complex analysis: Cauchy formula, argument principle, and Riemann mapping lectures.